Time Code Tool

Quick reference — reading, converting & generating timecode

A menu-bar timecode utility. **Read** LTC from any audio input or MTC from any MIDI port, and regenerate it as clean LTC audio and/or MTC — converting the frame rate on the way if you need to. Or run it as a **free-run master**, generating LTC and MTC together from one clock with no input at all. It also renders LTC audio files offline. Console-agnostic — useful to anyone who touches timecode.

Menu bar icon

Click	Show / hide the panel.
Right-click	Toggle the live timecode readout, or quit the app (no Dock icon or app menu — this is the only way to quit).
Icon colour	Grey = stopped or no signal. Green = locked, or generating. Amber = freewheel. Red = input clipping (overrides the others — fix the level first).
Live timecode text	Optional timecode in the menu bar, tinted by state (off by default; on a small display it can crowd the bar).

Choosing a mode

The switch at the top of the panel. Blue = reading, green = generating.

Read Timecode	Follow an incoming source: read LTC or MTC, show it, and regenerate it to the outputs.
Generate Timecode	Be the source: free-run as a timecode master from a start time and frame rate, with no input needed.
Start / Stop	Runs the mode you have selected. The mode, input source, and the generator's start time and frame rate are locked while running — stop first to change them. In 29.97 DF a start time at the top of a minute (00:17:00:00) doesn't exist — the app starts at 00:17:00;02 and tells you.

Input (Read Timecode)

LTC (audio)	Choose the interface carrying LTC and which channel to decode (mono). LTC often arrives on one channel of a multichannel device.
MTC (MIDI)	Read MIDI Time Code instead — from the app's own "Time Code Tool MTC In" port (select it as an MTC output in your DAW) or from any MIDI interface.
Readout	Large HH:MM:SS:FF; updates every frame while locked. Drop-frame shown with a ';'.
Frame rate	From LTC, auto-detected: 23.976 / 24 / 25 / 29.97 DF / 29.97 ND / 30. Shows "measuring…" briefly — telling 23.976 from 24 (and 29.97 ND from 30) needs a couple of seconds of signal.
Level meter	LTC input only. Distinguishes 'no LTC' from 'no cable', and warns when the feed is too hot — clipping distorts LTC and makes the readout jump.
Discontinuities	Counts frames that aren't previous + 1 (drop-frame aware), and says which kind: **Missed** (the signal path dropped frames; the timecode did not move), **Misread ignored** (an isolated frame with a bit error — the app drops it and carries on, no relatch, no MTC locate), **Jumps** (a real relocate, confirmed by the frame after it). Reset from the panel. If a source shows misreads, the app rides them out, but a clean source is still the fix — check a recording before the show with `tools/ltc_file_check.py`.

Important — MTC cannot distinguish 23.976 from 24, or 29.97 ND from 30. Timecode read over MTC therefore arrives labelled 24 or 30. Read it over LTC when that distinction matters.

Output

Both outputs are independent: either, both or neither. Each switch reveals its own settings.

Regenerate as MTC	Sends the timecode as MTC. Pick the destination: the app's own "Time Code Tool MTC Out" virtual port, or any MIDI interface, IAC bus or network session.
MTC rate	"Match input", or convert to a fixed rate on the way out. The panel shows the rate actually being transmitted.
Reading MTC?	MTC arrives with no sample clock of its own, so MTC output needs **Regenerate as LTC** switched on too — its audio device supplies the clock. The panel says so if it is missing.
Regenerate as LTC	Sends clean, freshly generated LTC audio to an output device — useful to reshape a ragged feed as well as to convert it.
Output device / Level	Any audio output, including the optional "Time Code Tool LTC" virtual device. Level in dBFS (default –18, ≈ 0 dBu at common calibration; range –24 to –3).
LTC rate	"Match input", or convert to a fixed rate — the same choice as MTC rate.
Freewheel	How many frames to keep generating through an input dropout before falling silent (default 5). Applies to both outputs.
Rescan devices	Re-scans audio and MIDI, so gear connected after launch appears without restarting.

Important — MTC cannot represent 23.976 or 29.97 ND. Those are sent under the nearest whole rate (24 and 30) and the panel shows a badge saying so. Every other rate is transmitted exactly, and LTC output has no such limit.

Generate Timecode (free-run master)

Start / Rate	The timecode to start from and the frame rate to generate (default 01:00:00:00 at 25).
Send MTC / Send LTC	Turn on either or both. Both run from one clock, so they stay locked to each other.
The clock	The audio output device provides the sample clock that paces both outputs. With Send LTC off, that device still runs as the clock — it just sends no LTC — so it stays selectable for MTC-only use.
Frame rate	Set once in the transport, so the per-output rate menus don't apply here.

Turn on at least one output — the app will not start with both switched off.

Virtual ports & devices

Time Code Tool MTC Out	A MIDI source the app creates. It stays available the whole time the app is running, so a DAW's routing survives stopping and restarting the transport.
Time Code Tool MTC In	A MIDI destination the app creates for reading MTC — select it as an MTC output in the sending application.
Time Code Tool LTC	An optional virtual audio device, offered as a tick-box when you run the installer — run it again later if you skipped it. With the device, other apps take this LTC as an audio input directly: no cabling, no spare interface channels. Needs macOS 13 or later; removal instructions are in Uninstalling.txt on the disk image.

Generate LTC File (offline)

Start / End / Duration	Start timecode, plus either an End timecode or a Duration in minutes (End takes precedence when set).
Rate / Sample rate / Bit depth	23.976–30 fps; 44.1 / 48 / 96 kHz; 16 / 24-bit. Mono.
Level	Output level in dBFS (default −18).
Render WAV	Offline render to a WAV you choose; the projected file size is shown first. WAV only — lossy formats smear the timecode square wave.

Running it

First launch	macOS prompts for microphone access the first time you read an audio input — allow it, or the app receives only silence (LTC on an input counts as ‘microphone’ to macOS).
Start at login	Optional, at the foot of the panel.
Decoder	LTC decoding uses libltc, bundled with the app.

If something goes wrong

Export log	Click **Export log** in the panel footer. A diagnostics zip lands on your Desktop and opens in Finder — this session’s log, the previous one, and your settings (licence details removed). Send that zip with your report; nothing leaves your Mac unless you send it.
What the log holds	Your settings at launch and each change, the audio and MIDI devices present, every Start with the mode, source and rates in force and every Stop, each lock, freewheel and locate, every error the panel showed you, your licence or trial state. Every launch is logged automatically.
The log folder	Right-click the menu bar icon and choose **Open Log Folder** (**~/Library/Logs/dylanmaudio/Time Code Tool/**, newest 20 launches kept).
Send feedback	Also in the footer — a short message straight to me, no email client needed.

On Windows

Install	Run **Time Code Tool Setup**: it installs for you alone, with no administrator prompt. Start menu → **dylanmaudio** holds the app, this guide and the uninstaller. An update installs over the old one; if Time Code Tool is running, Setup offers to close it first (say No if a show is running).
No LTC device	There's no Time Code Tool LTC device on Windows: choose your interface or a Dante channel as the LTC output. To pass LTC between apps on one PC, a virtual audio cable such as VB-CABLE works.
MTC ports	Windows can't make MIDI ports for an app. Create ports named **Time Code Tool MTC Out** and **Time Code Tool MTC In** in the free **loopMIDI**, and Time Code Tool uses them as its own; or choose any other MIDI port. The panel says which is missing.
MTC timing	On Windows, MTC runs about 45 ms (about a frame) behind the LTC from the same Time Code Tool, so its quarter-frames can go out evenly spaced. A receiver chasing MTC locks as usual; if you compare it against the LTC, allow for the offset.
Audio devices	The devices listed are Windows' own (WASAPI), once each. A Dante Virtual Soundcard channel shows as **DVS Transmit** or **DVS Receive** with its channel numbers. Time Code Tool opens a device at the rate it is already running at and never changes it.
Dante Virtual Soundcard	48 and 96 kHz both work. Whenever DVS's rate changes, set each DVS device's **Default Format** to the same rate (Sound settings → the device → Properties → Advanced). If they differ, Windows won't open the device, or the LTC reaches Time Code Tool too garbled to read.
Microphone access	Windows doesn't ask: Settings → Privacy & security → Microphone must have **Let desktop apps access your microphone** on, or LTC inputs read silence.
Remote Desktop	Connecting to the PC with Remote Desktop, disconnecting, or signing back in at it, restarts Windows' audio. Time Code Tool reopens its devices by itself and carries on from the timecode it was sending, but the timecode stops for a few seconds each time (up to about 10 while a connection is being made). Don't connect to a show PC while its timecode is running.
The tray icon	Windows puts a new tray icon under the **^** by the clock. Drag it onto the taskbar to keep it in view. Click and right-click work as the menu bar icon does above. Hover it for the running timecode (**Show timecode in tooltip** in its menu).
Licence	The same licence as on the Mac: its two activations can be on either, or one of each.
Logs and settings	Logs: **%LOCALAPPDATA%\dylanmaudio\Logs\Time Code Tool** (Open Log Folder in the tray menu). Settings: **%APPDATA%\dylanmaudio\Time Code Tool**. Export log saves its zip to your Desktop and opens Explorer.
Uninstall	Start menu → dylanmaudio → Uninstall Time Code Tool, or Settings → Apps. It asks whether to remove your settings and logs too. Your licence stays activated either way.

What's new in v1.4.2

• Fixed: MTC made from incoming LTC slowly ran ahead of the LTC it was reading: about one frame a minute at 30 fps and 48 kHz, half that at 96 kHz, building up until the timecode jumped or stopped. It now stays on the LTC for as long as the timecode runs. If you added a delay or offset downstream to make up for it, check it again with this version. Generating timecode and converting between frame rates were not affected.